MIL-I-38121B (USAF)
AMENDMENT 2
18 SEPTEMBER 1980
SUPERSEDING
AMENDMENT 1
24 March 1975

MILITARY SPECIFICATION

INDICATOR, PRESSURE EGU-9/A

This amendment forms a part of Military Specification MIL-I-38121B (USAF), dated 31 December 1970, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1, under SPECIFICATIONS, Federal: Delete "L-P-378" and "PPP-C-1120".

Add boxed paragraph to bottom center of page.

"Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter."

PAGE 3

3.5.3, second sentence: Delete and substitute the following: "The equipment lower test Mean-Time Between-Failure (MTBF), shall not be less than 1200 hours."

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4.5: Delete and substitute the following:
"4.5 Quality conformance inspection. The quality conformance tests identified in Table III shall be performed as described under paragraph 4.6 and consist of the following:

a. Individual tests
b. Sampling plans and tests"

4.5.1: Add the following:

"d. Position error
e. Integral lighting, when applicable
f. Dielectric strength"

FSC 6620

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PAGE 10

4.5.2.1: Delete: "f. Dielectric strength"

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4.5.2.2: Delete: "d. Torque".
Add:
"d. Extreme low temperature exposure
e. Extreme high temperature exposure
f. Salt spray
g. Reliability"

PAGE 14

4.6.15: Delete and substitite the following:
"4.6.15 Reliability. The indicator shall be subjected to a reliability demonstration at the following conditions of temperature, vibration, input voltage, and the cycling of these conditions to be applied to the equipment during the the test:

Temperature: -55°C to +85°C (-67°F to + 185°F)

Temperature Cycling: Temperature cycling shall be time to stabilize at low temperature followed by time to stabilize at the high temperature, plus two hours. Stabilize the equipment temperature at the lower test level temperature. Turn the equipment on and allow it to experience the combined effect of chamber heating and equipment operation until the point of maximum thermal inerita stabilizes at the higher temperature as determined in accordance with the thermal survey. The thermal survey shall have been previously conducted in accordance with paragraph 5.1.5 of MIL-STD-781. Maintain the equipment at the upper test level temperature for a period of two hours. Turn the equipment off and allow it to experience chamber cooling until the point of maximum thermal inertial stabilizes at the lower test level temperature. This cycle shall continue for the duration of the test.

Vibration: 2.0G±10% peak acceleration value at any nonresonant frequency between 20 and 60 hertz measured at the mounting points on the equipment. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time. When the equipment being tested contains circuit boards or cards, vibration shall be normal to the plane of the majority of the cards. Otherwise, the direction of vibration is not critical. This vibration requirement may be met by mounting the equipment solidly, (that is, without the equipment shock

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vibration mounts) to a strong flat plate which is supported by vibration isolators. The desired amplitide and frequency may be obtained by a simple vibration machine with an asymmetric weight on a shaft attached to the plate and driven by a suitable motor. It is recommended that the vibration equipment be checked for proper operation each 24 hours of operation, and that the vibration transducer be "on" and monitored continuously during vibration.

Equipment On-Off Cycling: Equipment off during cooling cycle and on during heating cycle.

Input Voltage: Nominal specified voltage +5 - 2%.

Input Voltage Cycling: Does not apply.

The accept-reject criteria shall be that of Test Plan IC of MIL-STD-781. The lower test MTBF(01) shall not be less than 1200 hours. The equipment shall be operated through the entire performance profile specified herein during the Reliability Demonstration Test."

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5: Delete 5. through 5.2.3 and substitute the following:

"5. Packaging

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified.

5.1.1 Level A.

5.1.1.1 Cleaning. Indicator shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Indicator shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4 Unit packaging. Unless otherwise specified by the procuring activity, each Indicator, pressure EGU-9/A, shall be packaged in quantity unit packs of one each in accordance with Method 1C-1 of MIL-P-116. Overbox in PPP-B-636 carton large enough to allow for application of sufficient cushioning material between container and bag, of a type, density, and

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thickness to insure shock transmission does not exceed peak values in Gs established for the item, when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Same as Level A except Method III of MIL-P-116 applies in lieu of specified method.

5.2 Packing. Packing shall be level A, B, or C as specified.

5.2.1 Level A. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, weather-resistant class, unless otherwise specified by the procuring activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, of minimum tare and cube consistent with the protection required. Table II of MIL-STD-794 applies for Army use.

5.2.2 Level B. Indicator packaged as specified in 5.1.1 shall be packed in domestic class exterior shipping containers conforming to PPP-B-636. Other requirements as specified in 5.2.1 above shall apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first domestic receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the contractor's commercial practice provided the latter meets the requirements of this level."

PAGE 17

6.4, last sentence: Delete and substitute:
"The activity responsible for the Qualified Products List is Hq AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016, and information pertaining to qualification of products may be obtained from that activity.

Add new paragraph 6.5
"6.5 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

Custodian:
Air Force - 99

Preparing Activity:
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